Sine, cosine, tangent and cotangent of an acute angle

Four ratios of sides that depend only on the angle — the whole of trigonometry starts here.

LESSON 19 | 1 × 40 MIN | GEOMETRY 8, TEMA 15 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Name the hypotenuse and the sides opposite and adjacent to a given acute angle.
- Define *sin*, *cos*, *tan*, *cot* as ratios of sides.
- Explain why the ratios depend on the angle and not on the size of the triangle.
- Calculate the four ratios from given side lengths.

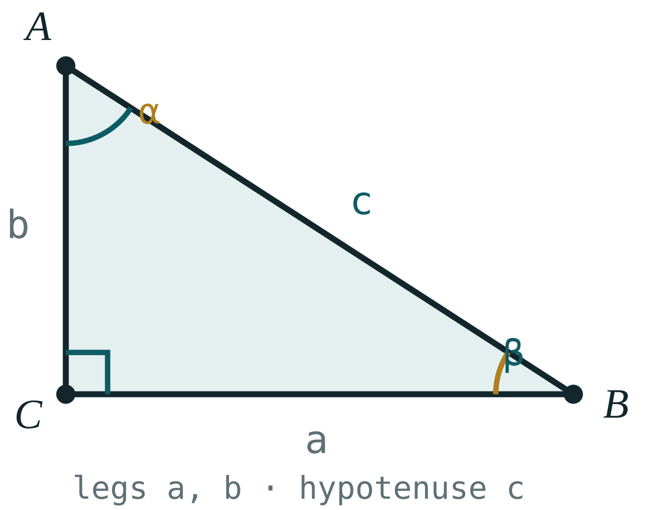
TEXTBOOK

National	Тема 15, pp. 35–37
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Naming the sides

In a right-angled triangle the side opposite the right angle is the **hypotenuse** — always the longest. The other two are the **legs**.



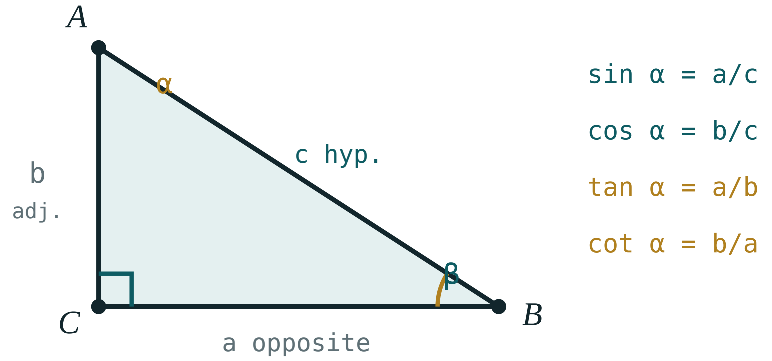
Right angle at C. The legs are a and b; the hypotenuse is c. Which leg is "opposite" depends on which acute angle you are looking at.

Fix your attention on the acute angle α at A. Then:

- $a = BC$ is the side **opposite** α — it does not touch A;
- $b = AC$ is the side **adjacent** to α — it does touch A, and it is not the hypotenuse;
- $c = AB$ is the **hypotenuse**.

> "OPPOSITE" MOVES
>
> Look at the other acute angle β and the roles swap: b becomes opposite and a becomes adjacent. The hypotenuse never changes.

The four definitions



The four ratios, all measured for the angle α at A.

$$\sin\alpha = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c} \qquad \cos\alpha = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}$$

$$\tan\alpha = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b} \qquad \cot\alpha = \frac{\text{adjacent}}{\text{opposite}} = \frac{b}{a}$$

Two immediate consequences, both worth saying aloud:

- $\tan\alpha \cdot \cot\alpha = 1$ — the two are reciprocals.
- $0 < \sin\alpha < 1$ and $0 < \cos\alpha < 1$, because a leg is always shorter than the hypotenuse. A sine of 1.2 is always an arithmetic error.

Why the ratios depend only on the angle

Draw two right-angled triangles with the same acute angle α. Their third angles are equal too (both $90° - \alpha$), so the triangles are **similar** — one is an enlargement of the other. Enlargement multiplies every side by the same scale factor, and in a ratio of two sides that factor cancels.

$$\frac{a}{c} = \frac{k \cdot a}{k \cdot c}$$

So $\sin\alpha$ is a property of the **angle**, not of the triangle. That is exactly why a single table of values can serve every right-angled triangle in the world.

02 Worked examples

EXAMPLE 1 ***In a right triangle $a = 3$, $b = 4$, $c = 5$. Find $\sin\alpha$, $\cos\alpha$, $\tan\alpha$ and $\cot\alpha$.***

1. $$\sin\alpha = \frac{a}{c} = \frac{3}{5} = 0.6$$

   Opposite over hypotenuse.

2. $$\cos\alpha = \frac{b}{c} = \frac{4}{5} = 0.8$$

   Adjacent over hypotenuse.

3. $$\tan\alpha = \frac{a}{b} = \frac{3}{4} = 0.75$$

   Opposite over adjacent.

4. $$\cot\alpha = \frac{b}{a} = \frac{4}{3}$$

   The reciprocal of the tangent.

ANSWER

$0.6,\ 0.8,\ 0.75,\ \frac{4}{3}$

EXAMPLE 2 ***In a right triangle $\sin\alpha = \frac{5}{13}$. Find $\cos\alpha$ and $\tan\alpha$.***

1. $opposite = 5,\ hypotenuse = 13$

   Read the ratio as two side lengths.

2. $b = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$

   Pythagoras gives the third side.

3. $\cos\alpha = \frac{12}{13}$

4. $\tan\alpha = \frac{5}{12}$

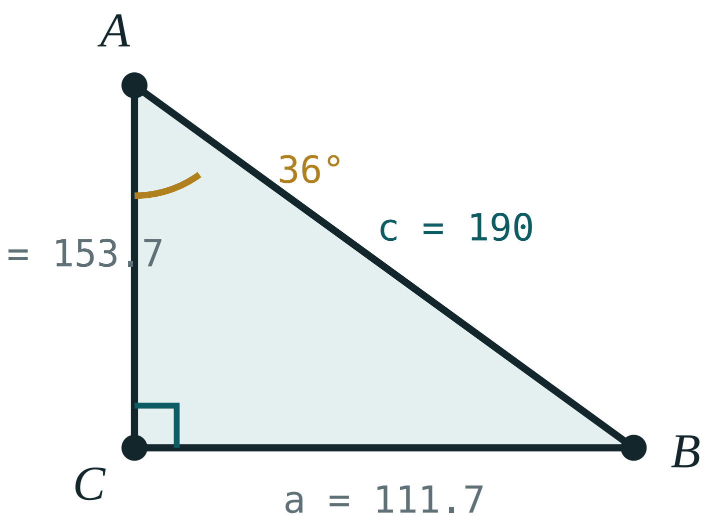
ANSWER

$\cos\alpha = \frac{12}{13},\ \tan\alpha = \frac{5}{12}$

03 Interactive model

Move the angle slider and watch the ratios change while the shape stays a right triangle.

Solve the right triangle



α **36°**	β **54°**	a (opposite) **111.7**	b (adjacent) **153.7**	c (hyp) **190**	sin α **0.59**
cos α **0.81**	tan α **0.73**	$a^2 + b^2$ **36100**	c^2 **36100**		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Right-angled triangle	To‘g‘ri burchakli uchburchak	Прямоугольный треугольник
Hypotenuse	Gipotenuza	Гипотенуза
Leg (cathetus)	Katet	Катет
Opposite side	Qarshi yotgan katet	Противолежащий катет
Adjacent side	Yondosh katet	Прилежащий катет
Sine	Sinus	Синус
Cosine	Kosinus	Косинус
Tangent	Tangens	Тангенс
Cotangent	Kotangens	Котангенс
Ratio	Nisbat	Отношение
Acute angle	O‘tkir burchak	Острый угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sin \alpha$ is:

- $\dfrac{\text{adjacent}}{\text{hypotenuse}}$
- $\dfrac{\text{opposite}}{\text{hypotenuse}}$
- $\dfrac{\text{opposite}}{\text{adjacent}}$
- $\dfrac{\text{hypotenuse}}{\text{opposite}}$

For an acute angle, $\sin \alpha$ is always:

- greater than 1
- between 0 and 1
- equal to 1
- negative

$\tan \alpha \cdot \cot \alpha$ equals:

- 0
- 1
- 2
- $\sin \alpha$

Two right triangles have the same acute angle. Their sines are:

- different
- equal
- reciprocal
- not comparable

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $a = 3, c = 5$. *Find* $\sin \alpha$.

   ANSWER 0.6

2. $b = 4, c = 5$. *Find* $\cos \alpha$.

   ANSWER 0.8

3. $a = 3, b = 4$. *Find* $\tan \alpha$.

   ANSWER 0.75

4. $a = 3, b = 4$. *Find* $\cot \alpha$.

   ANSWER $\frac{4}{3}$

5. *Which side is opposite the right angle?*

   ANSWER The hypotenuse.

6. *Can* $\sin \alpha = 1.2$ *for an acute angle?*

   ANSWER No — the sine of an acute angle is less than 1.

7. $a = 6, c = 10$. *Find* $\sin \alpha$.

   ANSWER 0.6

Medium · the standard question

7 PROBLEMS

1. $a = 5, b = 12, c = 13$. *Find all four ratios for* α.

ANSWER $sin = \frac{5}{13}, cos = \frac{12}{13}, tan = \frac{5}{12}, cot = \frac{12}{5}$

2. $\sin\alpha = \frac{5}{13}$. *Find* $\cos\alpha$.

ANSWER $\frac{12}{13}$

3. $\cos\alpha = \frac{8}{17}$. *Find* $\sin\alpha$.

ANSWER $\frac{15}{17}$

4. $\tan\alpha = \frac{3}{4}$. *Find* $\cot\alpha$.

ANSWER $\frac{4}{3}$

5. $a = 8, c = 17$. *Find* b *and* $\cos\alpha$.

ANSWER $b = 15, \cos\alpha = \frac{15}{17}$

6. *In the same triangle, write* $\sin\beta$ *in terms of the sides.*

ANSWER $\sin\beta = \frac{b}{c}$

7. $a = 9, b = 12$. *Find* c *and* $\sin\alpha$.

ANSWER $c = 15, \sin\alpha = 0.6$

Hard · stretch and reason

7 PROBLEMS

1. $\sin\alpha = \dfrac{7}{25}$. *Find* $\cos\alpha$, $\tan\alpha$ *and* $\cot\alpha$.

ANSWER $\cos = \dfrac{24}{25}$, $\tan = \dfrac{7}{24}$, $\cot = \dfrac{24}{7}$

2. $\tan\alpha = 1$. *What is* α, *and what are* $\sin\alpha$ *and* $\cos\alpha$?

ANSWER $\alpha = 45°$, both ratios $\dfrac{\sqrt{2}}{2}$

3. *Show that* $\sin\alpha = \cos\beta$ *in a right triangle with acute angles* α *and* β.

ANSWER Both equal $\dfrac{a}{c}$ for the same leg, since the two angles are complementary.

4. *A right triangle has legs in the ratio* $1 : 2$. *Find* $\tan\alpha$ *for the smaller angle.*

ANSWER $\tan\alpha = \dfrac{1}{2} = 0.5$

5. $c = 20$ *and* $\sin\alpha = 0.6$. *Find both legs.*

ANSWER $a = 12$, $b = 16$

6. *Prove that* $\tan\alpha = \dfrac{\sin\alpha}{\cos\alpha}$

ANSWER $\dfrac{a/c}{b/c} = \dfrac{a}{b} = \tan\alpha$

7. *A right triangle has* $\cot\alpha = 2.4$. *Find* $\tan\alpha$ *and the ratio of the legs.*

ANSWER $\tan\alpha = \dfrac{5}{12}$; legs in the ratio $5 : 12$

07 Homework

Homework — 5 problems

Geometry 8, Тема 15, pp. 35–37. Draw and label a triangle for every question.

1. $a = 7$, $b = 24$, $c = 25$. *Find the four ratios for* α.

2. *$\sin\alpha = 0.8$. Find $\cos\alpha$ and $\tan\alpha$.*
3. *$\cos\alpha = \frac{20}{29}$. Find $\sin\alpha$.*
4. *$a = 15$, $c = 17$. Find b, then $\tan\alpha$.*
5. *Explain in two sentences why $\sin\alpha$ does not depend on the size of the triangle.*